Universidad Internacional del Ecuador (UIDE)

Computer Science School ***Network Communication Technology***

Title: *Phase 2: Packet Tracer Network Prototype*

Professor: Steven Vinueza. M.Sc.

Team Members

Darwin Oswaldo Román Aguirre, 1726703760

Joselyn Paulet Laguna Puetate, 1753690617

José Emiliano Velasco Sosa, 1753931391

04 NOVEMBER, 2026

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1. Updated architecture

For this second phase of the project, we have adjusted and simplified part of the network architecture initially presented in Phase 1. The goal of these changes is to align the design with the practical concepts covered in class so far, facilitating its proper configuration and operation within the Cisco Packet Tracer simulator.

Justification for the changes made:

- **Connection between sites via serial cable:** Instead of using an intermediate wireless router—as originally proposed in the theoretical plan—we have decided to connect the router at Site A and the router at Site B directly using a serial cable. This is because wireless network configurations have not yet been covered in class; therefore, a serial cable allows us to simulate a direct, reliable point-to-point link between the two offices in a simpler way.
- **Use of the EIGRP routing protocol:** To enable communication between the two sites, we will implement the EIGRP protocol instead of OSPF. We chose this approach because EIGRP is the protocol we have most recently learned to configure, ensuring functional, dynamic routing between the networks.
- **Simplification of switch-to-switch connectivity:** For this initial phase, we have opted not to use EtherChannel or complex port redundancy protocols, as these topics have not yet been addressed in class. The connection between the switches at each site will be established directly using standard trunk port configuration (switchport mode trunk), ensuring that VLAN traffic is passed correctly, cleanly, and directly.

Identification of implemented and planned elements

Implemented elements (currently in the Packet Tracer prototype):

- Internal segmentation of each site using VLANs (VLAN 10 for users, VLAN 20 for servers, VLAN 30 for guests, and VLAN 99 for administration).
- Basic configuration of access and distribution switches, including the setup of trunk ports (switchport mode trunk).
- IP address allocation using a VLSM scheme, based on the corporate block 10.10.0.0/16.
- Configuration of DHCP services on the routers for the automatic assignment of IP addresses to user devices.
- Dynamic routing between Site A and Site B using the EIGRP protocol over a serial cable link.

Planned elements (for subsequent phases):

- Migration of dynamic routing from EIGRP to OSPF as the project integration progresses.
- Migration from simulated Cisco routers to physical devices running OpenWrt on Raspberry Pi boards (scheduled for week 10 onwards, once the physical hardware is delivered).
- Advanced implementation of link aggregation (EtherChannel) and loop-prevention protocols, such as RSTP, on the switches.
- Functional integration and final deployment of a Go-based chat server and a FastAPI authentication service within VLAN 20.

- Strict perimeter security measures, implemented via definitive Access Control Lists (ACLs/firewall rules) on the routers.

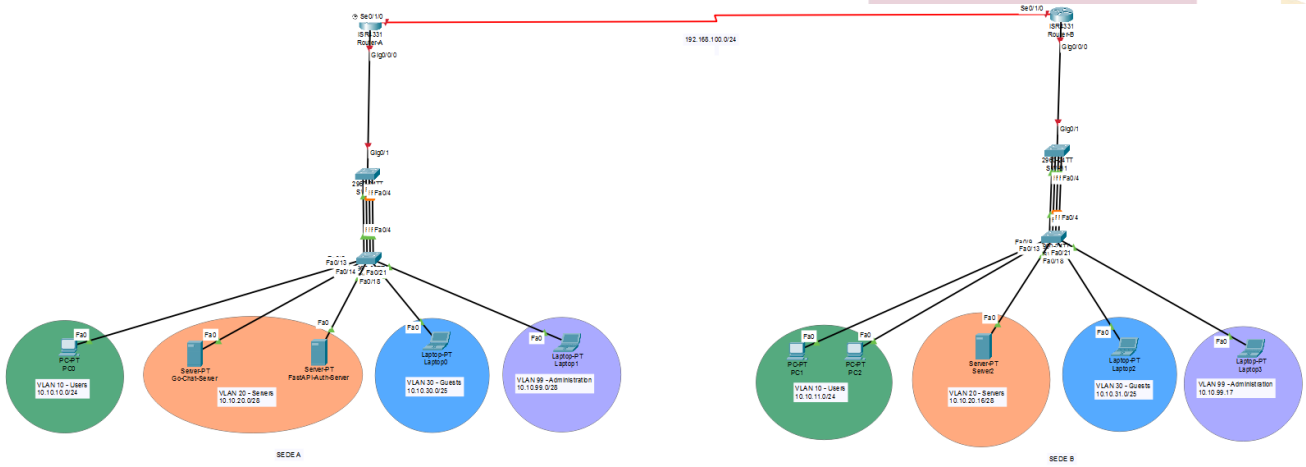
Site and connection labeling

In the updated Packet Tracer file, the topology is organized as follows:

- Site A: Includes the internal devices at the main office, such as the edge router (Router-A), distribution switches, and subinterfaces configured for VLANs 10, 20, 30, and 99.
- Site B: Comprises the remote branch office, including its respective switches and user equipment, managed via Router-B.
- Inter-site Serial Link: Directly connects the serial port of Router-A to the serial port of Router-B, enabling direct communication between both offices.

Figure 1

Logical topology in Cisco Packet Tracer for the enterprise network distributed between Site A and Site B



Note. Network infrastructure diagram in Cisco Packet Tracer showing the interconnection via a serial link (192.168.100.0/24) between Router-A and Router-B, logical segmentation via VLANs (VLAN 10, VLAN 20, VLAN 30, and VLAN 99) using VLSM subnets derived from 10.10.0.0/16, and the labeling of interfaces and end devices at both corporate locations.

Source: Created by the author (2026).

2. VLSM Addressing

Table 1

Purpose and calculation of VLSM subnets (Site A and Site B)

VLAN / Zone	Subnet Purpose	Host Requirements	Network & Prefix	Subnet Mask	Usable Range	Broadcast Address	Gateway Address	Allocation Type
VLAN 10 (Site A)	Workstations and users in the main office	Up to 254 hosts	10.10.10.0/24	255.255.255.0	10.10.10.1 - 10.10.10.254	10.10.10.255	10.10.10.254	**DHCP (Dynamic)**
VLAN 20	Central servers (Go-Chat and	Up to 14 hosts	10.10.20.0/28	255.255.255.240	10.10.20.1 - 10.10.20.14	10.10.20.15	10.10.20.14	**Static**

(Site A)	FastAPI-Auth)							
VLAN 30 (Site A)	Temporary network for guest devices	Up to 126 hosts	10.10.30.0/25	255.255.255.128	10.10.30.1 - 10.10.30.126	10.10.30.127	10.10.30.65	**DHCP (Dynamic)**
VLAN 99 (Site A)	Management and administration of network equipment (Site A)	Up to 14 hosts	10.10.99.0/28	255.255.255.240	10.10.99.1 - 10.10.99.14	10.10.99.15	10.10.99.10	**Static**
VLAN 10 (Site B)	Workstations and users in the remote Branch	Up to 254 hosts	10.10.11.0/24	255.255.255.0	10.10.11.1 - 10.10.11.254	10.10.11.255	10.10.11.100	**DHCP (Dynamic)**

VLAN 20 (Site B)	Support servers and branch nodes	Up to 14 hosts	10.10.20.16/28	255.255.255.240	10.10.20.17 - 10.10.20.30	10.10.20.31	10.10.20.30	**Static**
VLAN 30 (Site B)	Temporary network for branch guests	Up to 126 hosts	10.10.31.0/25	255.255.255.128	10.10.31.1 - 10.10.31.126	10.10.31.127	10.10.31.50	**DHCP (Dynamic)**
VLAN 99 (Site B)	Management and administration of network equipment (Site B)	Up to 14 hosts	10.10.99.16/28	255.255.255.240	10.10.99.17 - 10.10.99.30	10.10.99.31	10.10.99.25	**Static**
WAN Link	Point-to-point connection	2 hosts (Link)	192.168.100.0/24	255.255.255.0	192.168.100.1 - 192.168.100.2	192.168.100.255	N/A (Point-to-point)	**Static**

(Serial)	between Router-A and Router-B							

3. Basic device configuration

This section details the initial configuration applied to the network devices (routers and switches) at both sites. This includes assigning hostnames, configuring IP addressing for physical and virtual interfaces, saving configurations, and verifying the operational status of the equipment using console commands.

A. **Configuration of hostnames, interfaces, and addressing:** Each device was assigned a unique name to facilitate identification within the corporate topology. Additionally, the necessary network interfaces were configured:

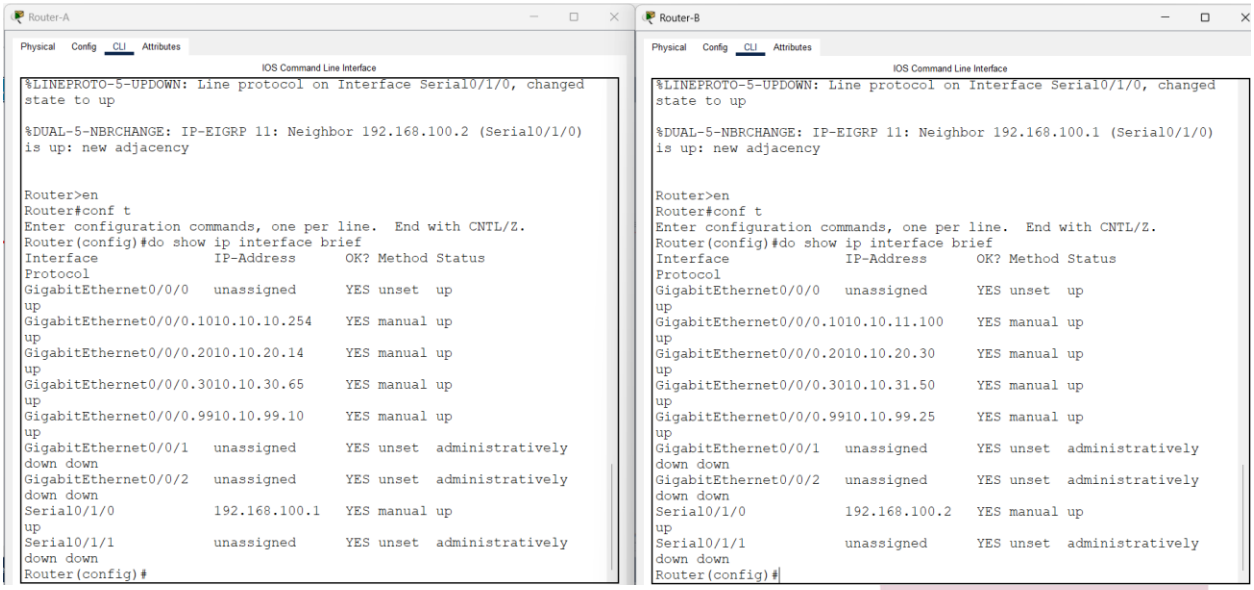
- **For the routers (Router-A and Router-B):** Subinterfaces associated with the VLANs were configured (using trunk encapsulation), along with the serial interface for the point-to-point link between sites, with IP addresses applied according to the VLSM (Variable Length Subnet Mask) plan.
- **For the switches (SW-A1, SW-A2, SW-B1, SW-B2):** Hostnames were assigned, VLANs (10, 20, 30, and 99) were formally created, and trunk ports (configured via `switchport mode trunk`) were set up to allow for the organized flow of traffic between devices.

- **Saving configurations:** The persistence command (write memory or copy running-config startup-config) was executed on all devices to ensure that changes are not lost upon system restart.

B. **Viewing active interfaces:** To verify that the physical and logical ports are operational and correctly linked (up/up), the interface verification command was used.

Figure 2

Verification of the operational status of active interfaces using the show ip interface brief` command



Note. Console output showing active physical interfaces and virtual subinterfaces, along with their respective assigned IP addresses and functional link states.

Source: Created by the author.

C. **Explanation of relevant entries in the MAC address table:** To ensure that switches at Site A and Site B can route traffic efficiently within the local network—without

saturating channels with unnecessary broadcasts—they dynamically build and update a MAC address table, also known as a CAM (Content Addressable Memory) table (GeeksforGeeks, 2025). This table maps the physical address (MAC address) of each device's network interface card to both the VLAN ID and the specific physical switch port through which the frame passed (Anything Over IP, 2026).

An analysis of the actual results obtained via verification commands on the distribution and access switches (SW-A1, SW-A2, SW-B1, and SW-B2) reveals the following key components:

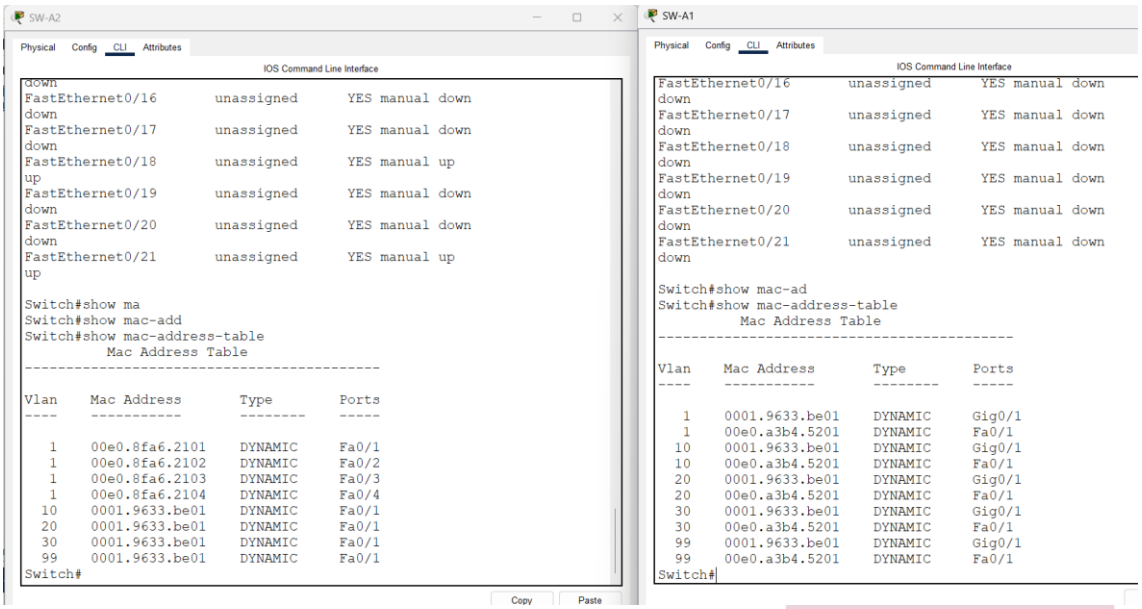
- **VLAN Association:** Table entries are explicitly segregated by their corresponding virtual network identifier (VLAN 1; VLAN 10 for users, VLAN 20 for servers, VLAN 30 for guests, and VLAN 99 for management). This ensures that Layer 2 traffic isolation remains intact across subnets (GeeksforGeeks, 2025).
- **Entry type (DYNAMIC):** All displayed addresses are categorized as "dynamic." This means the switch does not require manual pre-configuration; instead, it automatically learns the source MAC address of any incoming frame, immediately associates it with the corresponding port, and maintains an "aging timer" to update the network topology should a device move to a different port (Cisco Systems, 2026). Forwarding Ports: It can be observed how frames originating from end devices are mapped to specific access ports (e.g., Fa0/1 or Fa0/2), while inter-VLAN traffic directed toward routers or

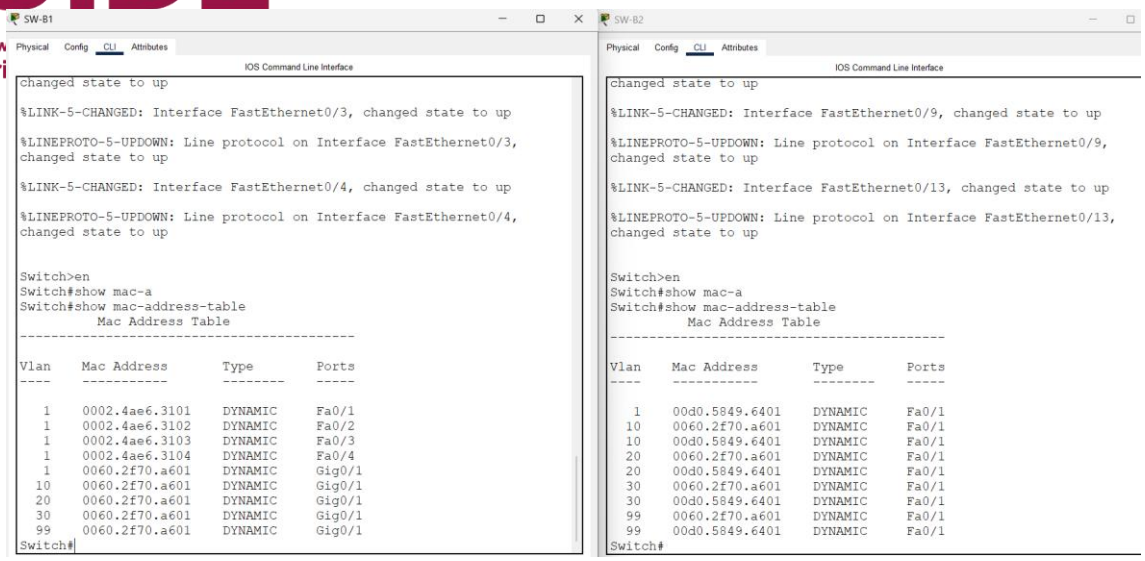
inter-switch links is efficiently aggregated via Gigabit trunk ports (Gig0/1) (Anything Over IP, 2026).

Through this intelligent learning and forwarding process, switches avoid blindly switching packets to every port, thereby enabling fast, orderly, and secure communication between the various subnets within the corporate platform (Cisco Systems, 2026).

Figure 3

Viewing the dynamic MAC address table on branch-site switches





Note. Console report obtained from Layer 2 switching devices, detailing the mapping between corporate VLANs, the MAC addresses of end devices, and the associated physical and trunk ports.

Source: Own work (2026).

4. DHCP

To automate the delivery of network parameters to workstations and user devices across the various subnets, DHCP was configured on the edge routers at each site (Router-A and Router-B). This allows end devices to dynamically obtain their IP addresses, subnet masks, and default gateways, in accordance with the ranges defined in the VLSM scheme.

A. Address pools (DHCP Pools) and gateways

Based on the planning from Phase 1, dynamic allocation pools were created for the active VLANs at both sites:

Site A:

- VLAN 10 (Users): 10.10.10.0/24 network configured with the gateway 10.10.10.254.

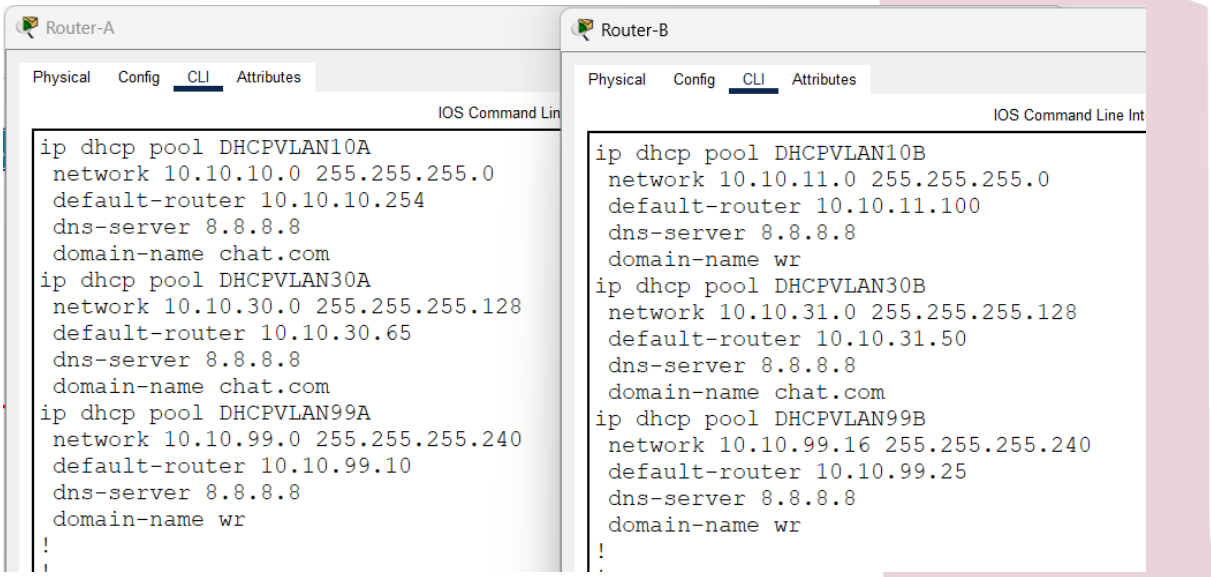
- VLAN 30 (Guests): 10.10.30.0/25 network configured with the gateway 10.10.30.65.
- VLAN 99 (Management): 10.10.99.0/28 network configured with the gateway 10.10.99.10.

Site B:

- VLAN 10 (Remote Users / Branch): 10.10.11.0/24 network configured with the gateway 10.10.11.100.
- VLAN 30 (Guests): 10.10.31.0/25 network configured with the gateway 10.10.31.50.
- VLAN 99 (Management): 10.10.99.16/28 network configured with the gateway 10.10.99.25.

Figure 4

Configuration of DHCP address pools and gateways on the edge routers



Note. Console view of the edge routers (Router-A and Router-B) showing the definition of DHCP pools for the various corporate VLANs and their respective assigned gateways.

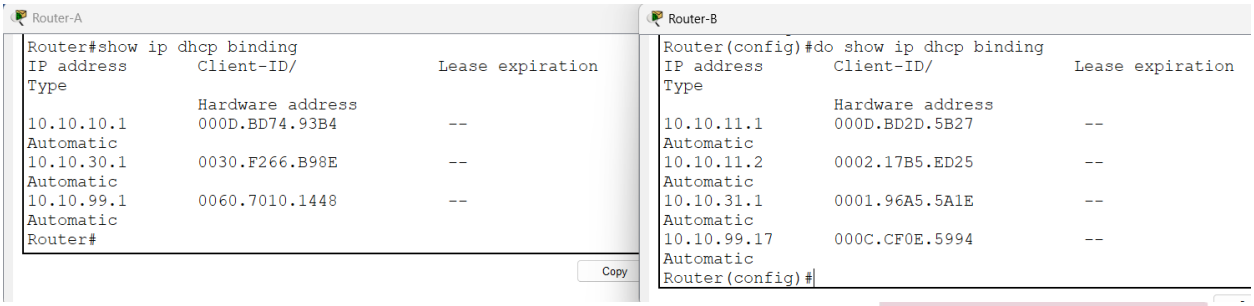
Source: Prepared by the autor.

B. Testing and verification of addresses received by clients

To confirm the correct delivery of parameters, the active lease table on the routers was verified using the appropriate control command, confirming that the connected devices obtained their IP addresses automatically, without any conflicts.

Figure 5

Verification of the DHCP lease table on the routers using the show ip dhcp binding command



Note. Console output detailing the IP addresses assigned dynamically (Automatic), along with the MAC addresses (hardware addresses) of the clients connected across the various subnets at Site A and Site B.

Source: Created by the autor.

5. Initial Inter-Site routing

To enable communication and packet exchange between Site A and Site B via a point-to-point serial link, the Enhanced Interior Gateway Routing Protocol (EIGRP) was implemented.

This dynamic routing protocol allows routers to automatically exchange information about their respective local networks.

A. Analysis of routing tables

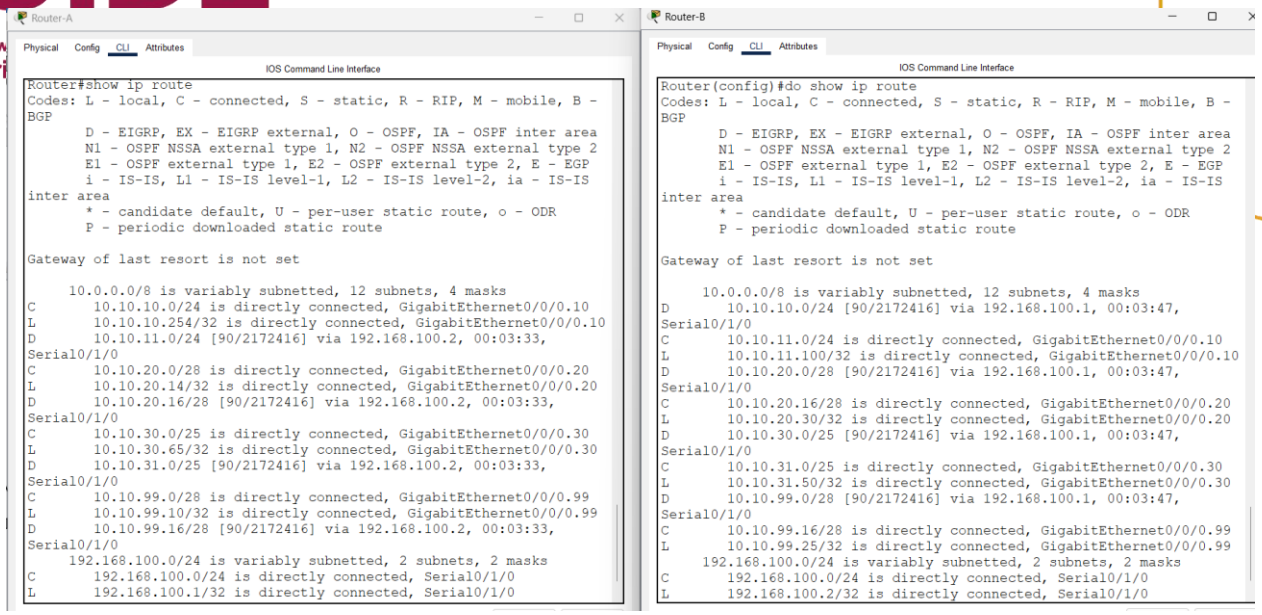
Executing the verification command show ip route on Router-A and Router-B clearly displays the learned routes:

Directly connected routes (C and L): These correspond to the local subnets at each site (VLANs 10, 20, 30, 99) and the serial WAN link (192.168.100.0/24) that physically connects both routers.

Routes learned via EIGRP (D): It can be observed that each router has automatically incorporated the subnets of the neighboring site using the EIGRP protocol, indicating their dynamic origin along with the associated metric via the serial interface.

Figure 6

Dynamic routing tables on the routers at both sites, displayed using the show ip route command



Note. Console output from the edge routers, showing both the directly connected local routes and the remote routes learned dynamically via the EIGRP protocol over the serial link.

Source: Created by the autor.

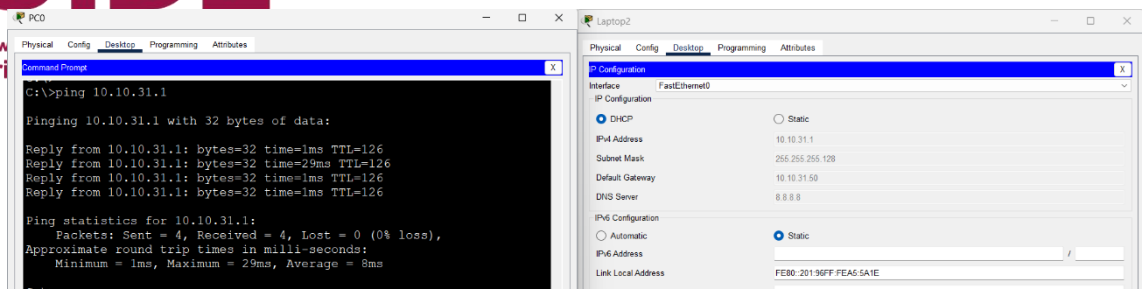
B. Connectivity and packet path testing between sites

To verify that dynamic routing is functioning correctly and that packets are traveling between offices without issues, connectivity tests (ping) were conducted from an end device at Site A to a device located at Site B.

The results demonstrate successful bidirectional communication, with 0% packet loss and optimal response times, confirming that traffic is being routed appropriately by the edge devices.

Figure 7

Successful connectivity test using the ping command between Site A and Site B



Note. Command window on a workstation at Site A executing an echo request (ping) to an address at Site B, confirming the effectiveness of inter-site routing.

Source: Created by the autor.

C. Justification for the temporary use of EIGRP and its replacement in later phases

The use of the EIGRP protocol in this second phase serves as an agile and functional prototyping mechanism within the Cisco Packet Tracer simulation. However, according to the project's comprehensive technical plan, this current routing scheme is only temporary.

For the subsequent stages (Phase 3), the plan includes migrating the routing infrastructure to the industry-standard OSPF (Open Shortest Path First) protocol. This will allow for the standardization of the corporate network architecture and its native integration with the physical, OpenWrt-based routers intended for the final system implementation.

6. Administration and troubleshooting

Incident: During the verification phase of dynamic routing between Router-A and Router-B, a temporary communication disruption occurred, preventing data packets from successfully traveling from one site to the other.

Diagnosis: A command was executed to check for neighboring devices on the network; it was discovered that some expected neighbors were missing, which caused the route to the other

branch to fail. Upon reviewing the active configuration, a minor typo was identified in the link parameters at one of the endpoints.

Corrective action: The configuration of the affected router was accessed, the relevant network parameter was corrected to align with the rest of the infrastructure, and the changes were saved.

Final verification (Re-test): Following the correction, the neighbor check was performed again, confirming that the connection was fully established. The complete restoration of traffic was then validated through echo (ping) tests between the various subnets.

7. Preliminary Tests

To ensure the proper operation of the implemented infrastructure, a battery of preliminary tests was executed to validate local network connectivity, automatic parameter assignment via DHCP, and inter-site communication. Each verification is detailed in the test log below:

Test ID	Source	Destination	Procedure	Expected Result	Observed Result	Evidence	Status
PR-01	Station PC0 (Site A, VLAN 10)	DHCP Server (Router-A)	Configure the client in DHCP mode to request	Automatic reception of a valid IP address, subnet mask,	The device dynamically obtained the IP address corresponding	Client IP configuration screenshot.	Passed

			network parameters.	and default gateway.	to the VLAN 10 pool.		
PR-02	Station PC1 / PC2 (Site B, VLAN 10)	DHCP Server (Router-B)	Request automatic configuration on the remote branch office devices.	Dynamic network assignment within the 10.10.11.0/24 segment.	Successful reception of network parameters from the Site B router.	Site B DHCP lease screenshot.	Passed
PR-03	Station PC0 (Site A)	Device in Site B (Laptop2)	Execute an echo command (ping) crossing the serial link and dynamic routes.	Successful response with 0% packet loss.	Bidirectional connectivity established with optimal response times and no losses.	Inter-site ping test.	Passed
PR-04	Workstation (Site A)	Advanced network servers	Request authentication services and	Active connection with the assigned	Functionality in the deployment process	N/A	Pending

			centralized data.	corporate servers.	according to the project's general schedule.		

8. Migration and completion Plan

To conclude the project and integrate the simulated infrastructure into a real production environment, based on equipment running the OpenWrt operating system, a migration plan for network functions (VLAN assignment, trunking, virtual interfaces, addressing, firewall services, and routing) is established.

A. Assignment of remaining network elements

VLANs and Trunking: In modern OpenWrt devices utilizing the Distributed Switch Architecture (DSA), the handling of tagged and untagged ports is managed through bridge-vlan configurations within the unified /etc/config/network file, replacing the traditional command-line syntax used for Cisco Catalyst switches (OpenWrt Project, 2026b).

Routing (Router-on-a-Stick and OSPF): Virtual subinterfaces on the router will be configured by creating logical devices based on the IEEE 802.1Q standard (for example, by associating VLAN suffixes with the base physical interface) (OpenWrt Project, 2026a). Regarding OSPF dynamic routing, unlike Cisco IOS, where it is configured directly via the operating system's CLI using global commands, OpenWrt requires the installation of additional routing packages based on specialized user-space daemons, such as Quagga (or Bird). These

utilize structured, plain-text configuration files similar to VTY (OpenWrt Project, 2023; TicArte, 2014). NAT and Firewall: While Cisco IOS uses access control lists (ACLs) and network address translation (NAT overload) commands, OpenWrt handles these security and masquerading functions natively via the Linux kernel's packet-filtering subsystem (nftables/iptables), managed through the Unified Configuration Interface (UCI) firewall configuration file (/etc/config/firewall) (OpenWrt Project, 2026a).

B. Mapping Cisco Concepts and Commands to OpenWrt

Proprietary commands from the Cisco IOS command-line interface (CLI) cannot be directly copied and pasted into the terminal of an OpenWrt device. Instead, OpenWrt employs an approach based on structured text configuration files known as the Unified Configuration Interface (UCI), which operates directly on native Linux kernel tools (such as iproute2, netifd`, and nftables) (OpenWrt Project, 2026a).

The conceptual and syntactic equivalents between the two environments are detailed below:

Network Concept / Function	**Traditional Cisco IOS CLI Syntax**	**Equivalent Implementation & Logic in OpenWrt (UCI / Linux)**	**Technical-Documentary Reference**
Interface Configuration & IP Addressing	interface GigabitEthernet0/0	Defined via logical interface blocks inside	(OpenWrt Project, 2026a)

	ip address 10.10.10.254 255.255.255.0	/etc/config/network specifying static protocol: config interface 'lan' option proto 'static' option ipaddr '10.10.10.254' option netmask '255.255.255.0'	
Subinterface Creation (Router-on-a-Stick)	interface GigabitEthernet0/0.10 encapsulation dot1Q 10 ip address 10.10.10.254 255.255.255.0	Implemented by configuring a virtual IEEE 802.1Q VLAN device in /etc/config/network: config device option type '8021q' option ifname 'eth0' option vid '10' option name 'eth0.10'	(OpenWrt Project, 2026b)
Dynamic Routing (OSPF Protocol)	router ospf 1 network 10.10.10.0 0.0.0.255 area 0	Managed by installing an external routing daemon (such as Quagga/OSPFd) configuring network blocks in	(OpenWrt Project, 2023; TicArte, 2014)

		its dedicated file /etc/quagga/ospfd.conf: router ospf network 10.10.10.0/24 area 0	
Network Address Translation (NAT Overload)	interface GigabitEthernet0/1 ip nat outside ip nat inside source list 1 interface overload	Configured directly in the rules and security zones subsystem of the /etc/config/firewall file, enabling masquerading: option masq '1'	(OpenWrt Project, 2026a)

9. Team Identification and Distribution of Contributions

This integrative project was developed collaboratively by the team, with responsibilities assigned according to each member's technical competencies:

- Joselyn Laguna: Responsible for the design and configuration of the network topology in Cisco Packet Tracer (implementation of VLANs, trunks, DHCP services, and EIGRP dynamic routing), as well as general technical writing and structuring the report in accordance with APA 7 standards.
- Darwin Román: Collaborated on the analysis of the IP addressing scheme using VLSM, the design of the subnet distribution table, and the verification of connectivity and preliminary testing.

- Jose Velasco: Participated in the theoretical research for the network migration, the structuring of device documentation, and the review of the transition plan toward architectures based on OpenWrt.

10. Link to the repository

https://github.com/Joselyn-uide/Chat-Empresarial-PI

11. Bibliographic references

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12. Disclosure Regarding the Use of Artificial Intelligence Assistance

Institutional transparency note: Please be advised that an artificial intelligence tool was used as a supplementary aid for reviewing the text, structuring technical sections, and formatting tables. The entire network configuration, the design within Cisco Packet Tracer, the validation of commands, and all engineering decisions were carried out and verified exclusively by the members of the project team.